

Numerical Examples: Electrostatics

Problem-1

Point charges 1 mC and -2 mC are located at $(3, 2, -1)$ and $(-1, -1, 4)$, respectively. Calculate the electric force on a 10 nC charge located at $(0, 3, 1)$ and the electric field intensity at that point.

Solution:

The force on a charge due to multiple point charges is given by

$$\mathbf{F} = \sum_{k=1}^2 \frac{QQ_k(\mathbf{r} - \mathbf{r}_k)}{4\pi\epsilon_0|\mathbf{r} - \mathbf{r}_k|^3} \quad (1)$$

Substituting the given values:

$$\mathbf{F} = \frac{Q}{4\pi\epsilon_0} \left[\frac{10^{-3}((0, 3, 1) - (3, 2, -1))}{|(0, 3, 1) - (3, 2, -1)|^3} - \frac{2 \times 10^{-3}((0, 3, 1) - (-1, -1, 4))}{|(0, 3, 1) - (-1, -1, 4)|^3} \right] \quad (2)$$

$$\mathbf{F} = \frac{10^{-3} \cdot 10 \times 10^{-9}}{4\pi\epsilon_0} \left[\frac{(-3, 1, 2)}{(9 + 1 + 4)^{3/2}} - \frac{2(1, 4, -3)}{(1 + 16 + 9)^{3/2}} \right] \quad (3)$$

$$\mathbf{F} = 9 \times 10^{-2} \left[\frac{(-3, 1, 2)}{14\sqrt{14}} + \frac{(-2, -8, 6)}{26\sqrt{26}} \right] \quad (4)$$

$$\mathbf{F} = -6.512 \mathbf{a}_x - 3.713 \mathbf{a}_y + 7.509 \mathbf{a}_z \text{ mN} \quad (5)$$

The electric field intensity is given by

$$\mathbf{E} = \frac{\mathbf{F}}{Q} \quad (6)$$

$$\mathbf{E} = (-6.512, -3.713, 7.509) \times \frac{10^{-3}}{10 \times 10^{-9}} \quad (7)$$

$$\mathbf{E} = -651.2 \mathbf{a}_x - 371.3 \mathbf{a}_y + 750.9 \mathbf{a}_z \text{ kV/m} \quad (8)$$

Problem2:

A circular ring of radius a carries a uniform charge ρ_L (C/m) and is placed on the xy -plane with its axis along the z -axis.

(a) Show that

$$\mathbf{E}(0, 0, h) = \frac{\rho_L a h}{2\epsilon_0 (h^2 + a^2)^{3/2}} \mathbf{a}_z \quad (9)$$

(b) What values of h give the maximum value of $\mathbf{E}$?

(c) If the total charge on the ring is Q , find $\mathbf{E}$ as $a \rightarrow 0$.

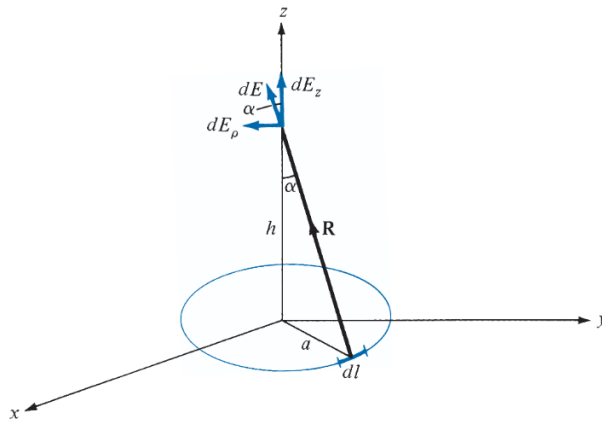


Figure 1: Circular Ring

Solution:

(a)

Consider a differential element on the ring:

$$dl = a d\phi \quad (10)$$

Position vector:

$$\mathbf{R} = -a \mathbf{a}_\rho + h \mathbf{a}_z \quad (11)$$

Magnitude:

$$R = |\mathbf{R}| = \sqrt{a^2 + h^2}, \quad \mathbf{a}_R = \frac{\mathbf{R}}{R} \quad (12)$$

Thus,

$$\frac{\mathbf{a}_R}{R^2} = \frac{\mathbf{R}}{|\mathbf{R}|^3} = \frac{-a \mathbf{a}_\rho + h \mathbf{a}_z}{(a^2 + h^2)^{3/2}} \quad (13)$$

Electric field:

$$\mathbf{E} = \frac{\rho_L}{4\pi\epsilon_0} \int_0^{2\pi} \frac{-a \mathbf{a}_\rho + h \mathbf{a}_z}{(a^2 + h^2)^{3/2}} a d\phi \quad (14)$$

By symmetry, $\mathbf{a}_\rho$ components cancel out, leaving only z -component:

$$\mathbf{E} = \frac{\rho_L a h}{4\pi\epsilon_0 (a^2 + h^2)^{3/2}} \int_0^{2\pi} d\phi \mathbf{a}_z \quad (15)$$

$$\mathbf{E} = \frac{\rho_L a h}{2\epsilon_0 (a^2 + h^2)^{3/2}} \mathbf{a}_z \quad (16)$$

(b)

Magnitude:

$$E = \frac{\rho_L a h}{2\epsilon_0 (h^2 + a^2)^{3/2}} \quad (17)$$

Differentiate w.r.t h :

$$\frac{dE}{dh} = 0 \quad (18)$$

This gives:

$$a^2 - 2h^2 = 0 \quad (19)$$

$$h = \pm \frac{a}{\sqrt{2}} \quad (20)$$

(c)

Total charge:

$$\rho_L = \frac{Q}{2\pi a} \quad (21)$$

Substitute:

$$\mathbf{E} = \frac{Q h}{4\pi\epsilon_0 (h^2 + a^2)^{3/2}} \mathbf{a}_z \quad (22)$$

As $a \rightarrow 0$:

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 h^2} \mathbf{a}_z \quad (23)$$

In general form:

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 r^2} \mathbf{a}_R \quad (24)$$

which is the field of a point charge.

Problem3

The finite sheet $0 \leq x \leq 1$, $0 \leq y \leq 1$ on the $z = 0$ plane has a charge density

$$\rho_s = xy(x^2 + y^2 + 25)^{3/2} \text{ nC/m}^2.$$

Find:

- (a) The total charge on the sheet
- (b) The electric field at $(0, 0, 5)$
- (c) The force experienced by a -1 mC charge located at $(0, 0, 5)$

Solution:

(a) Total Charge

$$Q = \iint_S \rho_s dS \tag{25}$$

$$= \int_0^1 \int_0^1 xy(x^2 + y^2 + 25)^{3/2} dx dy \text{ nC} \tag{26}$$

Let $u = x^2$, so $x dx = \frac{1}{2} du$:

$$Q = \frac{1}{2} \int_0^1 y \int_0^1 (x^2 + y^2 + 25)^{3/2} d(x^2) dy \tag{27}$$

$$= \frac{1}{2} \int_0^1 y \left[\frac{2}{5} (x^2 + y^2 + 25)^{5/2} \right]_0^1 dy \tag{28}$$

$$= \frac{1}{5} \int_0^1 y [(y^2 + 26)^{5/2} - (y^2 + 25)^{5/2}] dy \tag{29}$$

Let $v = y^2$, $y dy = \frac{1}{2} dv$:

$$Q = \frac{1}{10} \int_0^1 [(v + 26)^{5/2} - (v + 25)^{5/2}] dv \tag{30}$$

$$= \frac{1}{10} \cdot \frac{2}{7} [(v + 26)^{7/2} - (v + 25)^{7/2}]_0^1 \tag{31}$$

$$Q = \frac{1}{35} [(27)^{7/2} + (25)^{7/2} - 2(26)^{7/2}] \tag{32}$$

$$Q = 33.15 \text{ nC} \tag{33}$$

(b) Electric Field at $(0, 0, 5)$

$$\mathbf{E} = \iint_S \frac{\rho_s dS \mathbf{R}}{4\pi\epsilon_0 |\mathbf{R}|^3} \quad (34)$$

where

$$\mathbf{R} = (0, 0, 5) - (x, y, 0) = (-x, -y, 5)$$

$$\mathbf{E} = \int_0^1 \int_0^1 \frac{10^{-9} xy (x^2 + y^2 + 25)^{3/2} (-x \mathbf{a}_x - y \mathbf{a}_y + 5 \mathbf{a}_z)}{4\pi\epsilon_0 (x^2 + y^2 + 25)^{3/2}} dx dy \quad (35)$$

Using $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$:

$$\mathbf{E} = 9 \int_0^1 \int_0^1 xy (-x \mathbf{a}_x - y \mathbf{a}_y + 5 \mathbf{a}_z) dx dy \quad (36)$$

$$\mathbf{E} = 9 \left[- \int_0^1 x^2 dx \int_0^1 y dy \mathbf{a}_x - \int_0^1 x dx \int_0^1 y^2 dy \mathbf{a}_y + 5 \int_0^1 x dx \int_0^1 y dy \mathbf{a}_z \right] \quad (37)$$

$$\mathbf{E} = 9 \left(-\frac{1}{6}, -\frac{1}{6}, \frac{5}{4} \right) \quad (38)$$

$$\mathbf{E} = (-1.5, -1.5, 11.25) \text{ V/m} \quad (39)$$

(c) Force

$$\mathbf{F} = q\mathbf{E} \quad (40)$$

$$\mathbf{F} = (1.5, 1.5, -11.25) \text{ mN} \quad (41)$$

Problem 4

Planes $x = 2$ and $y = -3$, respectively, carry charges 10 nC/m^2 and 15 nC/m^2 . If the line $x = 0, z = 2$ carries charge $10\pi \text{ nC/m}$, calculate $\mathbf{E}$ at $(1, 1, -1)$ due to the three charge distributions.

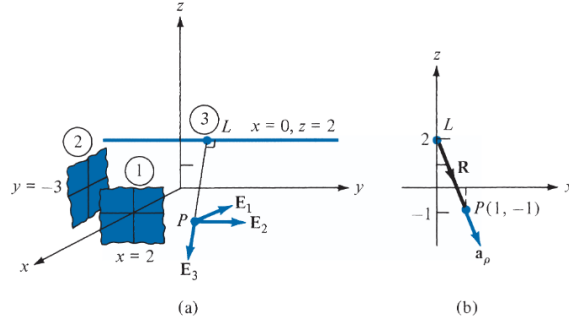


Figure 2: (a) three charge distributions, (b) finding ρ and $\mathbf{a}_\rho$ on plane $y = 1$.

Solution:

Let

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 + \mathbf{E}_3 \quad (42)$$

where $\mathbf{E}_1$, $\mathbf{E}_2$, and $\mathbf{E}_3$ are the contributions from the two infinite sheets and the line charge.

Field due to sheet at $x = 2$

$$\mathbf{E}_1 = \frac{\rho_s}{2\epsilon_0} (-\mathbf{a}_x) \quad (43)$$

$$= \frac{10 \times 10^{-9}}{2 \cdot \frac{10^{-9}}{36\pi}} \mathbf{a}_x \quad (44)$$

$$= -180\pi \mathbf{a}_x \quad (45)$$

Field due to sheet at $y = -3$

$$\mathbf{E}_2 = \frac{\rho_s}{2\epsilon_0} \mathbf{a}_y \quad (46)$$

$$= \frac{15 \times 10^{-9}}{2 \cdot \frac{10^{-9}}{36\pi}} \mathbf{a}_y \quad (47)$$

$$= 270\pi \mathbf{a}_y \quad (48)$$

Field due to line charge

$$\mathbf{E}_3 = \frac{\rho_L}{2\pi\epsilon_0\rho}\mathbf{a}_\rho \quad (49)$$

Distance vector from line to point:

$$\mathbf{R} = -3\mathbf{a}_z + \mathbf{a}_x \quad (50)$$

$$\rho = |\mathbf{R}| = \sqrt{10} \quad (51)$$

Unit vector:

$$\mathbf{a}_\rho = \frac{\mathbf{R}}{|\mathbf{R}|} = \frac{1}{\sqrt{10}}\mathbf{a}_x - \frac{3}{\sqrt{10}}\mathbf{a}_z \quad (52)$$

$$\mathbf{E}_3 = \frac{10\pi \times 10^{-9}}{2\pi \cdot \frac{10^{-9}}{36\pi}} \cdot \frac{1}{10}(\mathbf{a}_x - 3\mathbf{a}_z) \quad (53)$$

$$= 18\pi(\mathbf{a}_x - 3\mathbf{a}_z) \quad (54)$$

Total Electric Field

$$\mathbf{E} = (-180\pi + 18\pi)\mathbf{a}_x + 270\pi\mathbf{a}_y - 54\pi\mathbf{a}_z \quad (55)$$

$$= -162\pi\mathbf{a}_x + 270\pi\mathbf{a}_y - 54\pi\mathbf{a}_z \text{ V/m} \quad (56)$$

Problem 5

Determine $\mathbf{D}$ at $(4, 0, 3)$ if there is a point charge $-5\pi \text{ mC}$ at $(4, 0, 0)$ and a line charge $3\pi \text{ mC/m}$ along the y -axis.

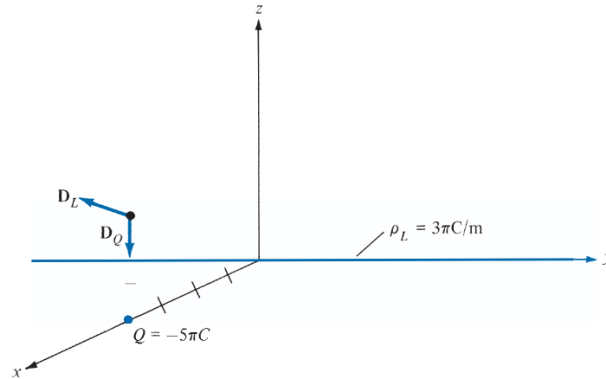


Figure 3: Flux density $\mathbf{D}$ due to a point charge and an infinite line charge

Solution:

Let

$$\mathbf{D} = \mathbf{D}_Q + \mathbf{D}_L \quad (57)$$

where $\mathbf{D}_Q$ and $\mathbf{D}_L$ are the flux densities due to the point charge and line charge, respectively.

Flux density due to point charge

$$\mathbf{D}_Q = \varepsilon_0 \mathbf{E} = \frac{Q}{4\pi R^2} \mathbf{a}_R = \frac{Q(\mathbf{r} - \mathbf{r}')}{4\pi |\mathbf{r} - \mathbf{r}'|^3} \quad (58)$$

$$\mathbf{r} - \mathbf{r}' = (4, 0, 3) - (4, 0, 0) = (0, 0, 3) \quad (59)$$

$$\mathbf{D}_Q = \frac{-5\pi \times 10^{-3}(0, 0, 3)}{4\pi(3)^3} \quad (60)$$

$$= -0.139 \mathbf{a}_z \text{ mC/m}^2 \quad (61)$$

Flux density due to line charge

$$\mathbf{D}_L = \frac{\rho_L}{2\pi\rho} \mathbf{a}_\rho \quad (62)$$

$$\mathbf{a}_\rho = \frac{(4, 0, 3) - (0, 0, 0)}{|(4, 0, 3) - (0, 0, 0)|} = \frac{(4, 0, 3)}{5} \quad (63)$$

$$\rho = |(4, 0, 3)| = 5 \quad (64)$$

$$\mathbf{D}_L = \frac{3\pi}{2\pi \cdot 5} \cdot \frac{(4\mathbf{a}_x + 3\mathbf{a}_z)}{5} \quad (65)$$

$$= \frac{3}{10} \cdot \frac{1}{5} (4\mathbf{a}_x + 3\mathbf{a}_z) \quad (66)$$

$$= 0.24 \mathbf{a}_x + 0.18 \mathbf{a}_z \text{ mC/m}^2 \quad (67)$$

Total Flux Density

$$\mathbf{D} = \mathbf{D}_Q + \mathbf{D}_L \quad (68)$$

$$= 0.24 \mathbf{a}_x + (0.18 - 0.139) \mathbf{a}_z \quad (69)$$

$$= 0.24 \mathbf{a}_x + 0.041 \mathbf{a}_z \text{ mC/m}^2 \quad (70)$$

$$\mathbf{D} = 240 \mathbf{a}_x + 41.1 \mathbf{a}_z \text{ } \mu\text{C/m}^2 \quad (71)$$

Problem 6

Given that

$$\mathbf{D} = z\rho \cos^2 \phi \mathbf{a}_z \text{ C/m}^2,$$

calculate the charge density at $(1, \pi/4, 3)$ and the total charge enclosed by the cylinder of radius 1 m with $-2 \leq z \leq 2$ m.

Solution:

The volume charge density is given by

$$\rho_v = \nabla \cdot \mathbf{D} = \frac{\partial D_z}{\partial z} \quad (72)$$

$$\rho_v = \rho \cos^2 \phi \quad (73)$$

At $(\rho, \phi, z) = (1, \pi/4, 3)$:

$$\rho_v = 1 \cdot \cos^2 \left(\frac{\pi}{4} \right) = \frac{1}{2} = 0.5 \text{ C/m}^3 \quad (74)$$

Total Charge

$$Q = \iiint_V \rho_v dv \quad (75)$$

$$= \iiint_V \rho \cos^2 \phi (\rho d\rho d\phi dz) \quad (76)$$

$$Q = \int_{-2}^2 dz \int_0^{2\pi} \cos^2 \phi d\phi \int_0^1 \rho^2 d\rho \quad (77)$$

Evaluate each integral:

$$\int_{-2}^2 dz = 4 \quad (78)$$

$$\int_0^{2\pi} \cos^2 \phi d\phi = \pi \quad (79)$$

$$\int_0^1 \rho^2 d\rho = \frac{1}{3} \quad (80)$$

$$Q = 4 \cdot \pi \cdot \frac{1}{3} \quad (81)$$

$$Q = \frac{4\pi}{3} \text{ C} \quad (82)$$

Problem 7

A charge distribution with spherical symmetry has density

$$\rho_v = \begin{cases} \frac{\rho_0 r}{R}, & 0 \leq r \leq R \\ 0, & r > R \end{cases}$$

Determine $\mathbf{E}$ everywhere.

Solution:

By symmetry, we apply Gauss's law:

$$\varepsilon_0 \oint_S \mathbf{E} \cdot d\mathbf{S} = Q_{\text{enc}} = \iiint_V \rho_v dv \quad (83)$$

(a) For $r < R$

$$\varepsilon_0 E_r 4\pi r^2 = Q_{\text{enc}} \quad (84)$$

$$Q_{\text{enc}} = \int_0^r \int_0^\pi \int_0^{2\pi} \frac{\rho_0 r'}{R} r'^2 \sin \theta d\phi d\theta dr' \quad (85)$$

$$= \int_0^r \frac{\rho_0 r'}{R} 4\pi r'^2 dr' \quad (86)$$

$$= \frac{4\pi\rho_0}{R} \int_0^r r'^3 dr' \quad (87)$$

$$= \frac{4\pi\rho_0}{R} \cdot \frac{r^4}{4} \quad (88)$$

$$Q_{\text{enc}} = \frac{\pi\rho_0 r^4}{R} \quad (89)$$

$$\varepsilon_0 E_r 4\pi r^2 = \frac{\pi\rho_0 r^4}{R} \quad (90)$$

$$\mathbf{E} = \frac{\rho_0 r^2}{4\varepsilon_0 R} \mathbf{a}_r \quad (91)$$

(b) For $r > R$

$$\varepsilon_0 E_r 4\pi r^2 = Q_{\text{total}} \quad (92)$$

$$Q_{\text{total}} = \int_0^R \frac{\rho_0 r'}{R} 4\pi r'^2 dr' \quad (93)$$

$$= \frac{4\pi\rho_0}{R} \int_0^R r'^3 dr' \quad (94)$$

$$= \frac{4\pi\rho_0}{R} \cdot \frac{R^4}{4} \quad (95)$$

$$Q_{\text{total}} = \pi\rho_0 R^3 \quad (96)$$

$$\varepsilon_0 E_r 4\pi r^2 = \pi\rho_0 R^3 \quad (97)$$

$$\mathbf{E} = \frac{\rho_0 R^3}{4\varepsilon_0 r^2} \mathbf{a}_r \quad (98)$$

Problem 8

Two point charges $-4\mu\text{C}$ and $5\mu\text{C}$ are located at $(2, -1, 3)$ and $(0, 4, -2)$, respectively. Find the potential at $(1, 0, 1)$, assuming zero potential at infinity.

Solution:

Let

$$Q_1 = -4\mu\text{C}, \quad Q_2 = 5\mu\text{C} \quad (99)$$

The potential is given by

$$V(\mathbf{r}) = \frac{Q_1}{4\pi\varepsilon_0|\mathbf{r} - \mathbf{r}_1|} + \frac{Q_2}{4\pi\varepsilon_0|\mathbf{r} - \mathbf{r}_2|} \quad (100)$$

Since $V(\infty) = 0$, the constant is zero.

Distances

$$|\mathbf{r} - \mathbf{r}_1| = |(1, 0, 1) - (2, -1, 3)| \quad (101)$$

$$= |(-1, 1, -2)| = \sqrt{6} \quad (102)$$

$$|\mathbf{r} - \mathbf{r}_2| = |(1, 0, 1) - (0, 4, -2)| \quad (103)$$

$$= |(1, -4, 3)| = \sqrt{26} \quad (104)$$

Potential Calculation

Using $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$:

$$V(1, 0, 1) = 9 \times 10^9 \left[\frac{-4 \times 10^{-6}}{\sqrt{6}} + \frac{5 \times 10^{-6}}{\sqrt{26}} \right] \quad (105)$$

$$V = 9 \times 10^3 \left[\frac{-4}{\sqrt{6}} + \frac{5}{\sqrt{26}} \right] \quad (106)$$

$$V = 9 \times 10^3(-1.633 + 0.9806) \quad (107)$$

$$V = -5.872 \text{ kV} \quad (108)$$

Problem 9

A point charge of 5 nC is located at $(-3, 4, 0)$, while the line $y = 1, z = 1$ carries a uniform charge 2 nC/m.

- (a) If $V = 0$ at $O(0, 0, 0)$, find V at $A(5, 0, 1)$.
- (b) If $V = 100 \text{ V}$ at $B(1, 2, 1)$, find V at $C(-2, 5, 3)$.
- (c) If $V = -5 \text{ V}$ at O , find V_{BC} .

Solution:

The total potential is

$$V = V_Q + V_L \quad (109)$$

Potential expressions

For the point charge:

$$V_Q = \frac{Q}{4\pi\epsilon_0 r} + C_1 \quad (110)$$

For the infinite line charge:

$$V_L = -\frac{\rho_L}{2\pi\epsilon_0} \ln \rho + C_2 \quad (111)$$

Thus,

$$V = -\frac{\rho_L}{2\pi\epsilon_0} \ln \rho + \frac{Q}{4\pi\epsilon_0 r} + C \quad (112)$$

where ρ is the perpendicular distance to the line and r is the distance from the point charge.

Distance definitions

$$\rho = \sqrt{(y-1)^2 + (z-1)^2} \quad (113)$$

(a) Potential at A

$$\rho_O = \sqrt{(0-1)^2 + (0-1)^2} = \sqrt{2}, \quad r_O = |(0,0,0) - (-3,4,0)| = 5 \quad (114)$$

$$\rho_A = \sqrt{(0-1)^2 + (1-1)^2} = 1, \quad r_A = |(5,0,1) - (-3,4,0)| = 9 \quad (115)$$

$$V_O - V_A = -\frac{\rho_L}{2\pi\epsilon_0} \ln \frac{\rho_O}{\rho_A} + \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r_O} - \frac{1}{r_A} \right) \quad (116)$$

Substitute values:

$$0 - V_A = -36 \ln \sqrt{2} + 45 \left(\frac{1}{5} - \frac{1}{9} \right) \quad (117)$$

$$V_A = 36 \ln \sqrt{2} - 4 = 8.477 \text{ V} \quad (118)$$

(b) Potential at C

$$\rho_B = 1, \quad r_B = \sqrt{21} \quad (119)$$

$$\rho_C = \sqrt{20}, \quad r_C = \sqrt{11} \quad (120)$$

$$V_C - V_B = -\frac{\rho_L}{2\pi\epsilon_0} \ln \frac{\rho_C}{\rho_B} + \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{r_C} - \frac{1}{r_B} \right) \quad (121)$$

$$V_C - 100 = -36 \ln \sqrt{20} + 45 \left(\frac{1}{\sqrt{11}} - \frac{1}{\sqrt{21}} \right) \quad (122)$$

$$V_C = 49.825 \text{ V} \quad (123)$$

(c) Potential difference

$$V_{BC} = V_C - V_B = 49.825 - 100 = -50.175 \text{ V} \quad (124)$$

Problem 10

Given the potential

$$V = \frac{10}{r^2} \sin \theta \cos \phi,$$

- (a) Find the electric flux density $\mathbf{D}$ at $(2, \pi/2, 0)$.
(b) Calculate the work done in moving a $10 \mu\text{C}$ charge from point $A(1, 30^\circ, 120^\circ)$ to $B(4, 90^\circ, 60^\circ)$.

Solution:

(a)

$$\mathbf{D} = \varepsilon_0 \mathbf{E} \quad (125)$$

Electric field:

$$\mathbf{E} = -\nabla V \quad (126)$$

In spherical coordinates,

$$\mathbf{E} = - \left[\frac{\partial V}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi \right] \quad (127)$$

Compute derivatives:

$$\frac{\partial V}{\partial r} = -\frac{20}{r^3} \sin \theta \cos \phi \quad (128)$$

$$\frac{\partial V}{\partial \theta} = \frac{10}{r^2} \cos \theta \cos \phi \quad (129)$$

$$\frac{\partial V}{\partial \phi} = -\frac{10}{r^2} \sin \theta \sin \phi \quad (130)$$

Thus,

$$\mathbf{E} = \frac{20}{r^3} \sin \theta \cos \phi \mathbf{a}_r - \frac{10}{r^3} \cos \theta \cos \phi \mathbf{a}_\theta + \frac{10}{r^3} \sin \theta \sin \phi \mathbf{a}_\phi \quad (131)$$

At $(r, \theta, \phi) = (2, \pi/2, 0)$:

$$\sin \theta = 1, \quad \cos \theta = 0, \quad \cos \phi = 1, \quad \sin \phi = 0$$

$$\mathbf{E} = \frac{20}{8} \mathbf{a}_r = 2.5 \mathbf{a}_r \quad (132)$$

$$\mathbf{D} = \varepsilon_0 \mathbf{E} = 2.5 \varepsilon_0 \mathbf{a}_r \quad (133)$$

$$\mathbf{D} = 22.1 \mathbf{a}_r \text{ pC/m}^2 \quad (134)$$

(b)

Work done:

$$W = -Q \int_A^B \mathbf{E} \cdot d\mathbf{l} = Q(V_B - V_A) \quad (135)$$

Evaluate potentials:

$$V_A = \frac{10}{1^2} \sin 30^\circ \cos 120^\circ = 10 \cdot \frac{1}{2} \cdot \left(-\frac{1}{2}\right) = -2.5 \quad (136)$$

$$V_B = \frac{10}{4^2} \sin 90^\circ \cos 60^\circ = \frac{10}{16} \cdot 1 \cdot \frac{1}{2} = \frac{10}{32} \quad (137)$$

$$W = 10 \times 10^{-6} \left(\frac{10}{32} - (-2.5) \right) \quad (138)$$

$$W = 10 \times 10^{-6} \left(\frac{10}{32} + \frac{5}{2} \right) \quad (139)$$

$$W = 28.125 \mu\text{J} \quad (140)$$

Problem 11

The point charges -1 nC , 4 nC , and 3 nC are located at $(0, 0, 0)$, $(0, 0, 1)$, and $(1, 0, 0)$, respectively. Find the energy in the system.

Solution:

The total electrostatic energy is

$$W = \frac{1}{2} \sum_{k=1}^3 Q_k V_k = \frac{1}{2} (Q_1 V_1 + Q_2 V_2 + Q_3 V_3) \quad (141)$$

Expanding:

$$W = \frac{Q_1}{2} \left(\frac{Q_2}{4\pi\epsilon_0(1)} + \frac{Q_3}{4\pi\epsilon_0(1)} \right) \quad (142)$$

$$+ \frac{Q_2}{2} \left(\frac{Q_1}{4\pi\epsilon_0(1)} + \frac{Q_3}{4\pi\epsilon_0(\sqrt{2})} \right) \quad (143)$$

$$+ \frac{Q_3}{2} \left(\frac{Q_1}{4\pi\epsilon_0(1)} + \frac{Q_2}{4\pi\epsilon_0(\sqrt{2})} \right) \quad (144)$$

Simplifying:

$$W = \frac{1}{4\pi\epsilon_0} \left(Q_1Q_2 + Q_1Q_3 + \frac{Q_2Q_3}{\sqrt{2}} \right) \quad (145)$$

Substitute $Q_1 = -1 \text{ nC}$, $Q_2 = 4 \text{ nC}$, $Q_3 = 3 \text{ nC}$:

$$W = \frac{1}{4\pi\epsilon_0} \left(-4 - 3 + \frac{12}{\sqrt{2}} \right) \times 10^{-18} \quad (146)$$

Using $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$:

$$W = 9 \left(\frac{12}{\sqrt{2}} - 7 \right) \text{ nJ} \quad (147)$$

$$W = 13.37 \text{ nJ} \quad (148)$$

Problem 12

A charge distribution with spherical symmetry has density

$$\rho_v = \begin{cases} \rho_0, & 0 \leq r \leq R \\ 0, & r > R \end{cases}$$

Determine V everywhere and the energy stored in region $r < R$.

Solution:

Using Gauss's law:

(a) For $r \geq R$

$$\mathbf{E} = \frac{\rho_0 R^3}{3\epsilon_0 r^2} \mathbf{a}_r \quad (149)$$

Potential:

$$V = - \int \mathbf{E} \cdot d\mathbf{l} = - \int \frac{\rho_0 R^3}{3\epsilon_0 r^2} dr \quad (150)$$

$$= \frac{\rho_0 R^3}{3\epsilon_0 r} + C_1 \quad (151)$$

Since $V(\infty) = 0$, $C_1 = 0$:

$$V = \frac{\rho_0 R^3}{3\epsilon_0 r}, \quad r \geq R \quad (152)$$

(b) For $r \leq R$

$$\mathbf{E} = \frac{\rho_0 r}{3\varepsilon_0} \mathbf{a}_r \quad (153)$$

$$V = - \int \frac{\rho_0 r}{3\varepsilon_0} dr = - \frac{\rho_0 r^2}{6\varepsilon_0} + C_2 \quad (154)$$

Continuity at $r = R$:

$$V(R) = \frac{\rho_0 R^2}{3\varepsilon_0} \quad (155)$$

$$\frac{\rho_0 R^2}{3\varepsilon_0} = - \frac{\rho_0 R^2}{6\varepsilon_0} + C_2 \quad (156)$$

$$C_2 = \frac{\rho_0 R^2}{2\varepsilon_0} \quad (157)$$

Thus,

$$V = \frac{\rho_0}{6\varepsilon_0} (3R^2 - r^2), \quad r \leq R \quad (158)$$

Final Result

$$V = \begin{cases} \frac{\rho_0 R^3}{3\varepsilon_0 r}, & r \geq R \\ \frac{\rho_0}{6\varepsilon_0} (3R^2 - r^2), & r \leq R \end{cases}$$

(c) Energy Stored for $r \leq R$

$$W = \frac{1}{2} \varepsilon_0 \int_V E^2 dv \quad (159)$$

For $r \leq R$:

$$E = \frac{\rho_0 r}{3\varepsilon_0} \quad (160)$$

$$W = \frac{1}{2} \varepsilon_0 \int_0^R \int_0^\pi \int_0^{2\pi} \left(\frac{\rho_0 r}{3\varepsilon_0} \right)^2 r^2 \sin \theta d\phi d\theta dr \quad (161)$$

$$W = \frac{1}{2} \varepsilon_0 \cdot \frac{\rho_0^2}{9\varepsilon_0^2} \int_0^R r^4 dr \cdot \int_0^\pi \sin \theta d\theta \cdot \int_0^{2\pi} d\phi \quad (162)$$

$$W = \frac{\rho_0^2}{18\varepsilon_0} \cdot \frac{R^5}{5} \cdot 2 \cdot 2\pi \quad (163)$$

$$W = \frac{2\pi \rho_0^2 R^5}{45\varepsilon_0} \text{ J} \quad (164)$$

Problem 13

Semi-infinite conducting planes at $\phi = 0$ and $\phi = \pi/6$ are separated by an infinitesimal insulating gap. If $V(\phi = 0) = 0$ and $V(\phi = \pi/6) = 100$ V, calculate V and $\mathbf{E}$ in the region between the planes.

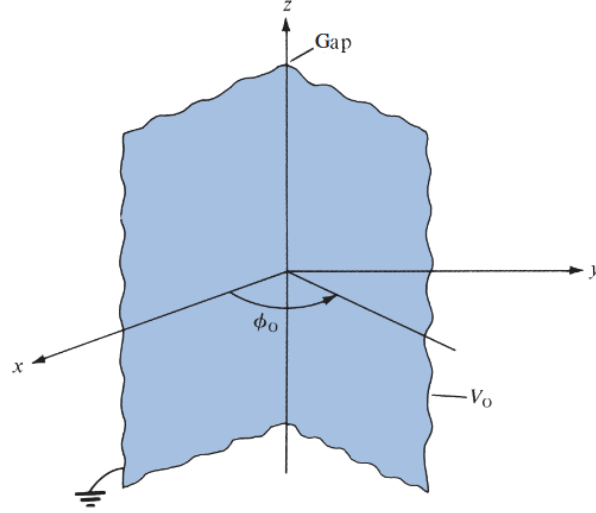


Figure 4: Potential $V(\phi)$ due to semi-infinite conducting planes

Solution:

Since V depends only on ϕ , Laplace's equation in cylindrical coordinates reduces to

$$\nabla^2 V = \frac{1}{\rho^2} \frac{d^2 V}{d\phi^2} = 0 \quad (165)$$

Multiplying by ρ^2 :

$$\frac{d^2 V}{d\phi^2} = 0 \quad (166)$$

Integrating twice:

$$V = A\phi + B \quad (167)$$

Applying Boundary Conditions

At $\phi = 0$, $V = 0$:

$$B = 0 \quad (168)$$

At $\phi = \phi_0$, $V = V_0$:

$$V_0 = A\phi_0 \quad \Rightarrow \quad A = \frac{V_0}{\phi_0} \quad (169)$$

Thus,

$$V = \frac{V_0}{\phi_0} \phi \quad (170)$$

Electric Field

$$\mathbf{E} = -\nabla V = -\frac{1}{\rho} \frac{dV}{d\phi} \mathbf{a}_\phi \quad (171)$$

$$\mathbf{E} = -\frac{V_0}{\rho \phi_0} \mathbf{a}_\phi \quad (172)$$

Final Result

Substitute $V_0 = 100$ and $\phi_0 = \pi/6$:

$$V = \frac{600}{\pi} \phi \quad (173)$$

$$\mathbf{E} = -\frac{600}{\pi \rho} \mathbf{a}_\phi \quad (174)$$

Check

$$\nabla^2 V = 0, \quad V(0) = 0, \quad V\left(\frac{\pi}{6}\right) = 100 \quad (175)$$

Problem 14

Determine the capacitance of each of the capacitors in Figure 5. Take $\varepsilon_{r1} = 4$, $\varepsilon_{r2} = 6$, $d = 5 \text{ mm}$, $S = 30 \text{ cm}^2$.

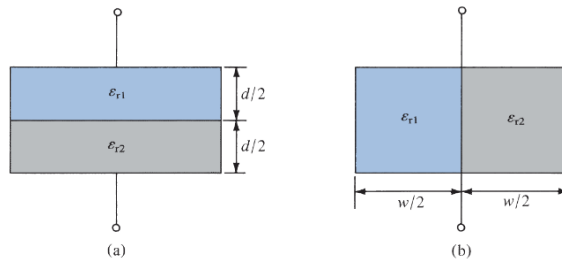


Figure 5: Capacitors

Solution:

(a) Series configuration

Since $\mathbf{D}$ and $\mathbf{E}$ are normal to the dielectric interface, the capacitor can be treated as two capacitors in series.

$$C_1 = \frac{\varepsilon_0 \varepsilon_{r1} S}{d/2} = \frac{2\varepsilon_0 \varepsilon_{r1} S}{d}, \quad C_2 = \frac{2\varepsilon_0 \varepsilon_{r2} S}{d} \quad (176)$$

Total capacitance:

$$C = \frac{C_1 C_2}{C_1 + C_2} = \frac{2\varepsilon_0 S}{d} \cdot \frac{\varepsilon_{r1} \varepsilon_{r2}}{\varepsilon_{r1} + \varepsilon_{r2}} \quad (177)$$

Substituting values:

$$C = 2 \cdot \frac{10^{-9}}{36\pi} \cdot \frac{30 \times 10^{-4}}{5 \times 10^{-3}} \cdot \frac{4 \times 6}{10} \quad (178)$$

$$C = 25.46 \text{ pF} \quad (179)$$

(b) Parallel configuration

Here $\mathbf{D}$ and $\mathbf{E}$ are parallel to the interface, so the capacitors are in parallel.

$$C_1 = \frac{\varepsilon_0 \varepsilon_{r1} (S/2)}{d} = \frac{\varepsilon_0 \varepsilon_{r1} S}{2d}, \quad C_2 = \frac{\varepsilon_0 \varepsilon_{r2} S}{2d} \quad (180)$$

Total capacitance:

$$C = C_1 + C_2 = \frac{\varepsilon_0 S}{2d} (\varepsilon_{r1} + \varepsilon_{r2}) \quad (181)$$

Substituting values:

$$C = \frac{10^{-9}}{36\pi} \cdot \frac{30 \times 10^{-4}}{2 \cdot (5 \times 10^{-3})} \cdot 10 \quad (182)$$

$$C = 26.53 \text{ pF} \quad (183)$$

Problem 15

A cylindrical capacitor has radii $a = 1 \text{ cm}$ and $b = 2.5 \text{ cm}$. If the space between the plates is filled with an inhomogeneous dielectric with

$$\varepsilon_r = \frac{10 + \rho}{\rho},$$

where ρ is in centimeters, find the capacitance per meter of the capacitor.

Solution:

The potential difference is given by

$$V = - \int_b^a \frac{Q}{2\pi\epsilon_0\epsilon_r\rho L} d\rho \quad (184)$$

$$= - \frac{Q}{2\pi\epsilon_0 L} \int_b^a \frac{d\rho}{\rho \left(\frac{10+\rho}{\rho} \right)} \quad (185)$$

$$V = - \frac{Q}{2\pi\epsilon_0 L} \int_b^a \frac{d\rho}{10 + \rho} \quad (186)$$

$$V = - \frac{Q}{2\pi\epsilon_0 L} \ln(10 + \rho) \Big|_b^a \quad (187)$$

$$V = \frac{Q}{2\pi\epsilon_0 L} \ln \left(\frac{10 + b}{10 + a} \right) \quad (188)$$

Capacitance per unit length

$$C = \frac{Q}{V} = \frac{2\pi\epsilon_0}{\ln \left(\frac{10+b}{10+a} \right)} \quad (189)$$

Substitute values ($L = 1$ m, $\epsilon_0 = \frac{10^{-9}}{36\pi}$):

$$C = 2\pi \cdot \frac{10^{-9}}{36\pi} \cdot \frac{1}{\ln \left(\frac{12.5}{11.0} \right)} \quad (190)$$

$$C = 434.6 \text{ pF/m} \quad (191)$$

Problem 16

Two extensive homogeneous isotropic dielectrics meet on plane $z = 0$. For $z > 0$, $\epsilon_{r1} = 4$ and for $z < 0$, $\epsilon_{r2} = 3$. A uniform electric field

$$\mathbf{E}_1 = 5\mathbf{a}_x - 2\mathbf{a}_y + 3\mathbf{a}_z \text{ kV/m}$$

exists for $z \geq 0$.

Find:

- (a) $\mathbf{E}_2$ for $z \leq 0$
- (b) The angles $\mathbf{E}_1$ and $\mathbf{E}_2$ make with the interface

- (c) The energy densities (in J/m³) in both dielectrics
- (d) The energy within a cube of side 2 m centered at (3, 4, -5)

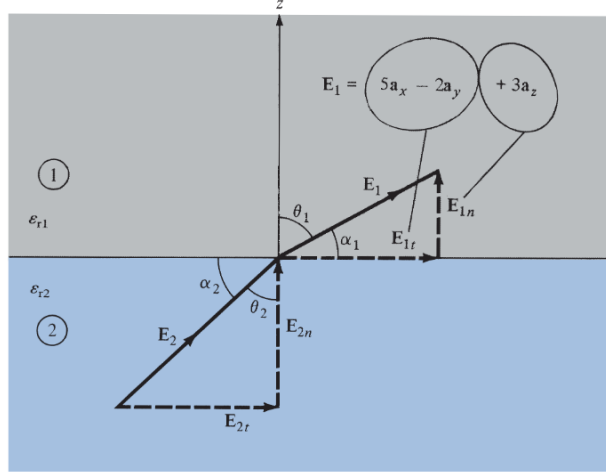


Figure 6: Dielectric Boundary

Solution:

(a) Field in region 2

Normal vector: $\mathbf{a}_n = \mathbf{a}_z$

$$\mathbf{E}_{1n} = (\mathbf{E}_1 \cdot \mathbf{a}_z)\mathbf{a}_z = 3\mathbf{a}_z \quad (192)$$

$$\mathbf{E}_{1t} = \mathbf{E}_1 - \mathbf{E}_{1n} = 5\mathbf{a}_x - 2\mathbf{a}_y \quad (193)$$

Tangential continuity:

$$\mathbf{E}_{2t} = \mathbf{E}_{1t} = 5\mathbf{a}_x - 2\mathbf{a}_y \quad (194)$$

Normal boundary condition:

$$\varepsilon_1 E_{1n} = \varepsilon_2 E_{2n} \quad (195)$$

$$E_{2n} = \frac{\varepsilon_{r1}}{\varepsilon_{r2}} E_{1n} = \frac{4}{3} (3\mathbf{a}_z) = 4\mathbf{a}_z \quad (196)$$

$$\mathbf{E}_2 = 5\mathbf{a}_x - 2\mathbf{a}_y + 4\mathbf{a}_z \text{ kV/m} \quad (197)$$

(b) Angles with interface

$$E_{1t} = \sqrt{5^2 + (-2)^2} = \sqrt{29}, \quad E_{1n} = 3 \quad (198)$$

$$\tan \theta_1 = \frac{E_{1t}}{E_{1n}} = \frac{\sqrt{29}}{3} \Rightarrow \theta_1 = 60.9^\circ \quad (199)$$

Angle with interface:

$$\alpha_1 = 90^\circ - \theta_1 = 29.1^\circ \quad (200)$$

Similarly,

$$E_{2t} = \sqrt{29}, \quad E_{2n} = 4 \quad (201)$$

$$\tan \theta_2 = \frac{\sqrt{29}}{4} \Rightarrow \theta_2 = 53.4^\circ \quad (202)$$

$$\alpha_2 = 90^\circ - \theta_2 = 36.6^\circ \quad (203)$$

(c) Energy densities

$$w_{E1} = \frac{1}{2} \varepsilon_1 |\mathbf{E}_1|^2 = \frac{1}{2} (4\varepsilon_0) (25 + 4 + 9) \times 10^6 \quad (204)$$

$$= 672 \mu\text{J}/\text{m}^3 \quad (205)$$

$$w_{E2} = \frac{1}{2} \varepsilon_2 |\mathbf{E}_2|^2 = \frac{1}{2} (3\varepsilon_0) (25 + 4 + 16) \times 10^6 \quad (206)$$

$$= 597 \mu\text{J}/\text{m}^3 \quad (207)$$

(d) Energy in cube

The cube lies entirely in region 2 ($z < 0$). Volume:

$$V = 2 \times 2 \times 2 = 8 \text{ m}^3$$

$$W = w_{E2} \cdot V = 597 \times 8 \mu\text{J} = 4.776 \text{ mJ} \quad (208)$$

Problem 17

Region $y < 0$ consists of a perfect conductor while region $y > 0$ is a dielectric medium ($\varepsilon_r = 2$). If there is a surface charge of $2 \text{ nC}/\text{m}^2$ on the conductor, determine $\mathbf{E}$ and $\mathbf{D}$ at:

- (a) $A(3, -2, 2)$

(b) $B(-4, 1, 5)$

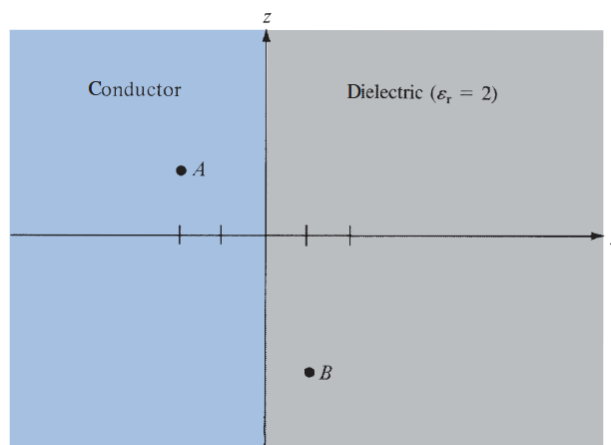


Figure 7: Perfect conductor- Dielectric

Solution:

(a) Point $A(3, -2, 2)$

Since $y = -2 < 0$, point A lies inside the conductor. Hence,

$$\mathbf{E} = 0, \quad \mathbf{D} = 0 \quad (209)$$

(b) Point $B(-4, 1, 5)$

Since $y = 1 > 0$, point B lies in the dielectric.

Normal component of $\mathbf{D}$:

$$D_n = \rho_s = 2 \text{ nC/m}^2 \quad (210)$$

Direction is along $\mathbf{a}_y$, hence:

$$\mathbf{D} = 2 \mathbf{a}_y \text{ nC/m}^2 \quad (211)$$

Electric field:

$$\mathbf{E} = \frac{\mathbf{D}}{\epsilon_0 \epsilon_r} \quad (212)$$

$$\mathbf{E} = \frac{2 \times 10^{-9}}{\left(\frac{10^{-9}}{36\pi}\right) \cdot 2} \mathbf{a}_y \quad (213)$$

$$= 36\pi \mathbf{a}_y \text{ V/m} \quad (214)$$

$$\mathbf{E} = 113.1 \mathbf{a}_y \text{ V/m} \quad (215)$$